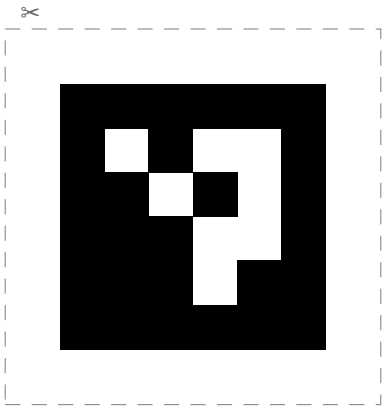
Table reference markers

A4 | Actual size / 100% | Turn off Fit to page

Print in black and white, single sided. Do not use borderless scaling.

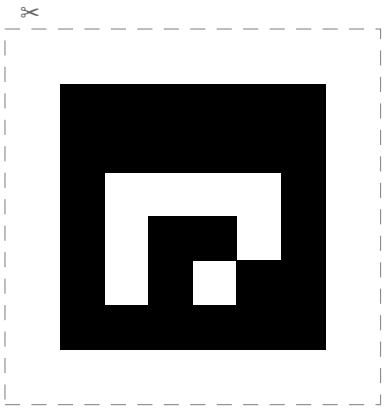
ArUco DICT_4X4_50. Cut out and place around the working area.

ID 0 - Table reference



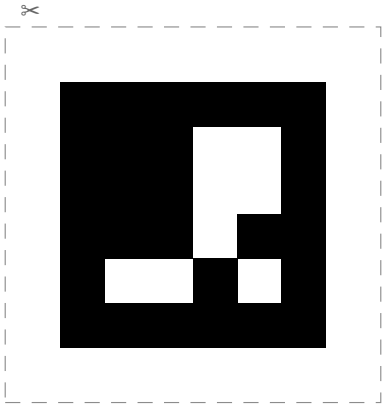
Black square: 35 x 35 mm

ID 1 - Table reference



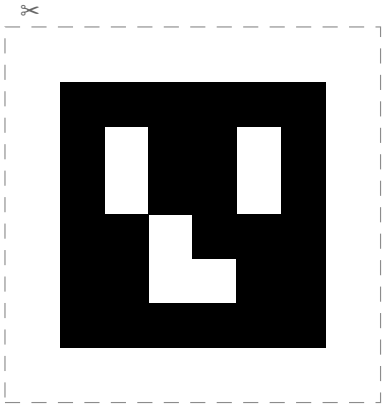
Black square: 35 x 35 mm

ID 2 - Table reference



Black square: 35 x 35 mm

ID 3 - Table reference

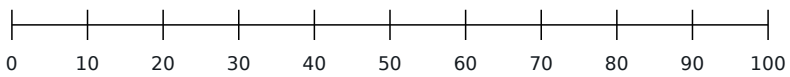


Black square: 35 x 35 mm

Cut along the dashed grey lines. Keep the white border inside. Keep markers flat.

This page is not a table layout. Measure the marker centre positions on the table.

Print check: this line must measure exactly 100 mm



After printing, measure the black squares and check line with a physical ruler.

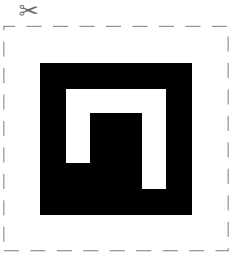
Hand and object markers

A4 | Actual size / 100% | Turn off Fit to page

Print in black and white, single sided. Do not use borderless scaling.

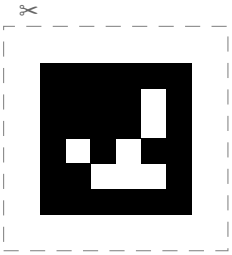
Use IDs 10 and 11 for simple plate pushing. IDs 12 and 13 are optional.

ID 10 - Hand / contact



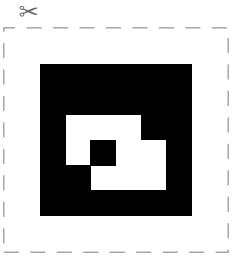
Black square: 20 x 20 mm

ID 11 - Plate / object



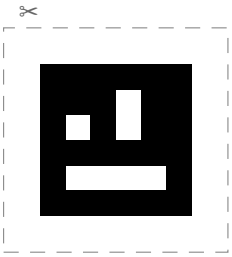
Black square: 20 x 20 mm

ID 12 - Finger / spare



Black square: 20 x 20 mm

ID 13 - Finger / spare



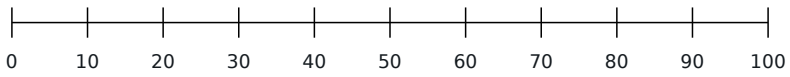
Black square: 20 x 20 mm

ID 10: fix to a flat, visible surface on the hand. Measure its offset to the contact point.

ID 11: place on the flat centre of the plate. Keep it visible during the push.

IDs 12/13: spare finger markers. Finger opening tracking is not used in this pipeline.

Print check: this line must measure exactly 100 mm



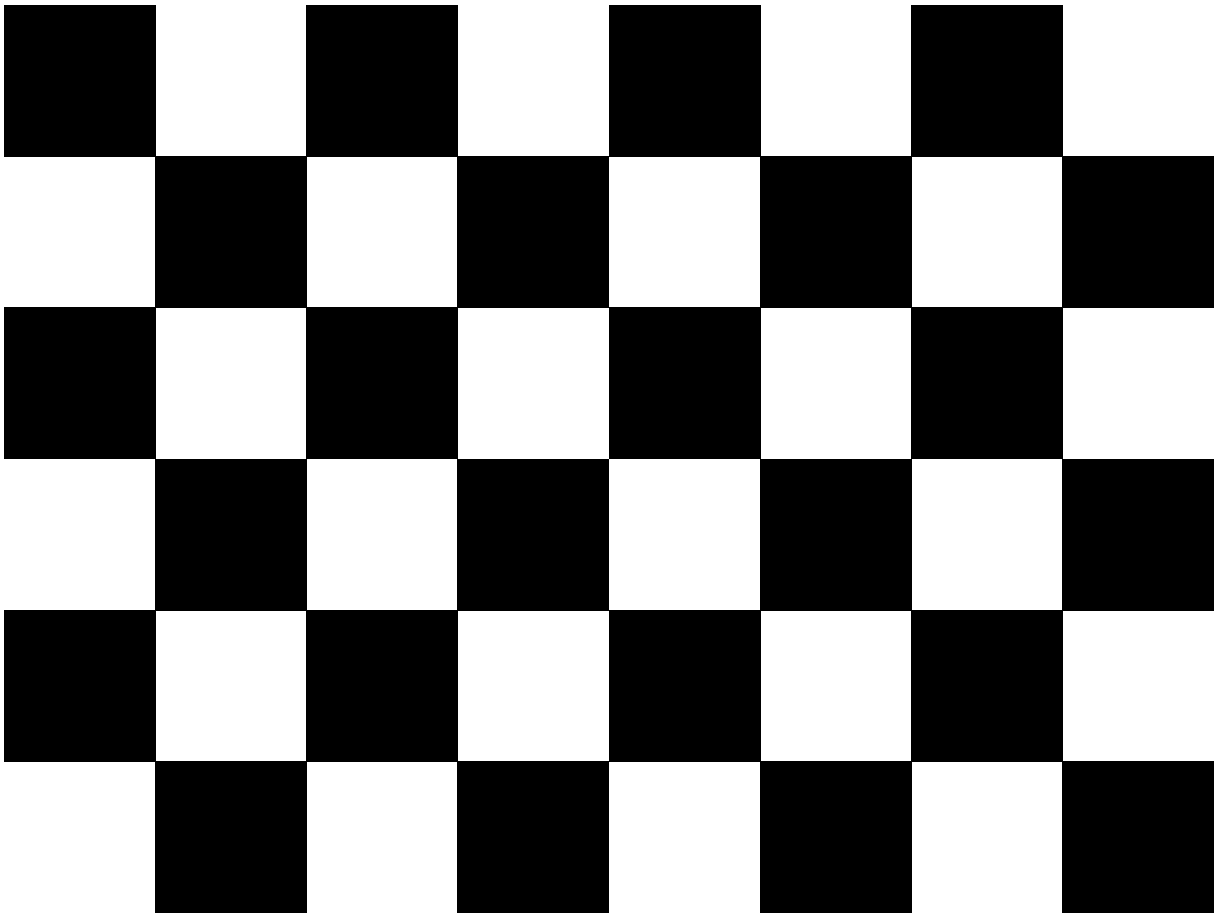
Marker size includes the black outer border, and excludes the surrounding white area.

Camera calibration board

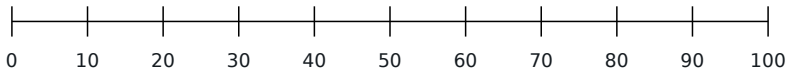
A4 | Actual size / 100% | Turn off Fit to page

Print in black and white, single sided. Do not use borderless scaling.

8 x 6 squares; 7 x 5 inner corners. Each square: 20 x 20 mm. Board: 160 x 120 mm.



Print check: this line must measure exactly 100 mm



Mount on a flat, rigid surface. Calibrate with the same lens, zoom and video settings.

Printing these pages does not calibrate the camera. Measure the printed dimensions.

Check one short pilot recording before collecting the full demonstration videos.